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Title: A PbWO₄ Scintillator Range-Telescope for Real-Time Depth-Dose Measurement in Low-Dose Environments for Proton Therapy

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Editor report

1 Summary of the Manuscript

The manuscript presents a study on a segmented PbWO₄ scintillator detector designed for real-time depth-dose and range measurements of proton beams in low-dose environments. By utilizing Silicon Photomultipliers (SiPMs) and the high density and low light yield of PbWO₄, the authors develop a compact device intended to avoid signal saturation in the Bragg peak region. The methodology includes custom electronics and a Bayesian unfolding method to account for photon statistics and SiPM microcell saturation. The results demonstrate sub-millimeter WET (water-equivalent thickness) accuracy using a 221.6 MeV proton beam.

2 General Evaluation

The manuscript describes a functional prototype and a rigorous approach to signal processing. The detector addresses a specific application for low-dose, single-particle tracking, such as pCT and Monte Carlo validation.

I recommend publication after major revisions. I recommend that the authors address gaps regarding the calibration strategy, WET calculations, and the physical definition of the measured dose before the manuscript is suitable for publication.

3 Major Comments

3.1. Definition of "Dose" vs. "Primary Range Distribution"

In Section 2.3.4 (Lines 349–355), the authors state that 50% to 70% of all events are rejected because they represent protons that scattered out of the small detector cross-section or underwent nuclear interactions.

- Rejecting these scattered protons implies the detector is measuring the idealized single-particle dE/dx track of the primary beam, rather than a macroscopic clinical depth-dose distribution where scatter and secondary particles contribute to the total absorbed dose.
- I recommend to explicitly state in the text that the resulting curves represent the range distribution of unscattered primaries. This distinction is necessary to prevent confusion with standard clinical dosimetry measurements.

3.2. Energy Calibration Logic

In Section 2.3.1, the rationale for using Geant4 and ^{60}Co calibrations for different channels is contradictory as currently written.

- Line 212 states the thin crystals have the highest light yield, but Line 242 states their low light yield makes gamma-source measurements challenging.
- Revise Lines 241–247 to clarify this logic. Explicitly state that channels 30–32 must be calibrated with a gamma source because they are located beyond the Bragg peak (preventing the use of the Geant4 "stopped proton" method). Clarify that their higher intrinsic light yield is what makes the ^{60}Co calibration physically possible despite their thin geometry.

3.3. WET Simulation Approximation

In Section 2.3.2 (Lines 321–326), the beam energy for the WET simulation is calculated as ~ 195 MeV based on the 52 mm PMMA phantom. The authors state that for simplicity, this same energy approximation is used for all phantoms.

- The manuscript claims sub-millimeter precision (deviations $< 200 \mu\text{m}$ in Table 1). Because stopping power is energy-dependent, using a static 195 MeV beam to simulate WET for thicker phantoms (which reduce the incident energy further) introduces a systematic error.
- The authors should either re-run the Geant4 simulations with the correct estimated energies for the 53–57 mm phantoms or provide quantitative evidence that the error introduced by this approximation is negligible compared to the stated sub-millimeter deviations.

3.4. Missing SiPM Parameter

In Section 2.3.1 (Equations 2.10 and 2.11), the derivation utilizes a Binomial distribution to account for SiPM microcell saturation.

- The parameter N_{cells} is required for this calculation, but the numerical value for the selected AFBR-S4N44P014M SiPM is omitted.
- I would recommend providing the total number of microcells (N_{cells}) in the text to allow the reader to reproduce the detector response function (DRF).

3.5. Sensitivity to Birks' Constant Uncertainty

In Section 2.3.1 (Lines 261–262), the authors assume a 30% relative uncertainty for Birks' coefficient (k_B) due to a lack of literature values. Quenching is highly significant in the Bragg peak region.

- Add a statement in the Results/Discussion section detailing how sensitive the final Bragg peak position (R_0) and amplitude are to this $\pm 30\%$ variation, and state whether it dominates the systematic error budget.

4 Minor Comments

1. **Line 13:** "is an under utilized" should be "are an underutilized".
2. **Line 14:** "Additionally useful for detector development..." is a sentence fragment. Revise to: "These measurements are additionally useful for...".
3. **Line 16:** "they are by unsuited" should be revised to "making them unsuited" for grammatical flow.

4. **Lines 51-52:** "measurements in low-dose regimes remain largely under-utilized, offering access to single particle tracking. These regimes enable access to single-particle information..." This is repetitive. Condense to avoid restating the access to single-particle tracking.
5. **Line 54:** Insert "such as" before "proton Computed Tomography" and use the standard abbreviation "pCT" instead of "PCT".
6. **Line 75:** "wedge on scintillator systems" should be hyphenated as "wedge-on-scintillator systems".
7. **Line 80:** Remove the unnecessary comma after "While": "While all these devices..."
8. **Line 128:** Grammatical agreement error: "The amplifier supply as well as the SiPM bias voltages are..." Change "are" to "is", or rewrite as "Both the amplifier supply and the SiPM bias voltages are..."
9. **Line 131:** Replace the period with a comma: "...pitch size of 40 μm , making them sensitive..."
10. **Line 153:** "The resulting combination... result in" should be "results in".
11. **Line 160:** Italicize the variable k ("number of photoelectrons k ") to match the equation formatting.
12. **Line 229:** Missing period: "...particularly of the SiPM Laboratory tests showed..." should be "...of the SiPM. Laboratory tests showed..."
13. **Line 344:** Specify whether the "50 ns coincidence window" is applied globally across all 32 channels or only between adjacent channels.
14. **Line 366:** Punctuation error: "...rate dependencies of the electronics such as dead time and pile-up, can in principle..." Either remove the comma after 'electronics' or add one before 'such as'.
15. **Line 385:** Add a brief reminder that ϵ represents the tail of non-elastic interactions, explicitly connecting it to the event rejection described in Section 2.3.4.
16. **Line 448:** Grammatical error: "The total deposited energy, given by the product of the mean energy loss and the number of protons and was found to be..." Remove "and" so it reads "...number of protons, was found to be..."

17. **Line 448:** Missing period and space: " 2×10^{-6} J For the..." should be " 2×10^{-6} J. For the...".
18. **Line 449:** Typo and missing period: " 1×10^{-7} J The..." should be " 1×10^{-7} J. The...".
19. **Figure 3:** The scatter point markers for the measured data are too small to differentiate from the analytical lines in the zoomed-in view (b). Increase the marker size for legibility.